

MATIC NUMBER									

DC014
Sains Komputer 1
Semester I
Sesi 2024/ 2025
2 jam

NAME: _____ PRACTICUM: _____

KOLEJ MATRIKULASI JOHOR
KEMENTERIAN PENDIDIKAN MALAYSIA
JOHOR MATRICULATION COLEGE
MISTRY OF EDUCATION MALAYSIA

PRA PSPM 1
PROGRAM MATRIKULASI SISTEM EMPAT SEMESTER
PRE -PSPM 1
MATRICULATION PROGRAMME FOUR SEMESTER SYSTEM

SAINS KOMPUTER 1
2 JAM

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON:

Kertas ini mengandungi 6 soalan.
Jawab semua soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE: This booklet consists of 6 questions. Answer all questions in the space provided.

UNTUK KEGUNAAN PEPERIKSA		
No. Soalan	Pemeriksa	KKP/KP
1		
2		
3		
4		
5		
6		
Jumlah		

Kertas soalan ini mengandungi 8 halaman bercetak, termasuk muka hadapan.
This booklet consists 8 printed pages, include cover page.

1. Miss Rozaini, is a science teacher. She teaches about the habitat of tigers by using multimedia presentation in giving information to the student.

- a) Name two (2) **most suitable multimedia element** that Miss Rozaini can apply in her lesson. [2 marks]

- b) Describe one (1) purpose of using the multimedia element mentioned in question 1 a). [2 marks]

- c) She prefers bitmap images instead of vector image as elements in her presentation. Describe the disadvantage of using vector image. [2 marks]

2. Multimedia development is driven by a project-based approach, and a well-structured plan can help conserve both time and resources. The creation of interactive multimedia applications is a complex task that requires several stages of development.

- a) Identify the phase based on the activities. [6 marks]

Scenario	Phase
(i) The content of a multimedia contents will be delivered online, and the students will access it on their smartphones, tablets, and laptops.	
(ii) You are hired to shoot and edit a highlight video for a corporate event. The video needs to showcase key moments from a conference and be ready for publishing online within 48 hours.	
(iii) The budget for the entire multimedia project is RM10,000, and the company has a tight timeline of 4 weeks to complete the production. You need to ensure the production efficiently to stay within budget and meet the deadline.	
(iv) A multimedia e-commerce website is to be developed for a new fashion brand for a specific target audience, and the site should reflect a modern, user-friendly design.	
(v) Part of the activities in multimedia development is to listen particularly to the vocal level in the audio used and the consistency of the sound between tracks.	
(vi) The script of a video demonstration project has been written by the marketing team and includes voiceover text, on-screen text, and a call-to-action. You are to ensure the script is clear, concise, and free of any errors or inconsistencies before it moves into the production phase.	

- b) Multimedia authoring tools allow users to create, organize, and edit multimedia projects by combining text, images, audio, and video, making it easier to produce interactive or multimedia-rich content.
- i) Determine which authoring tool type best fits the descriptions provided.

[2 marks]

Description	Type authoring tool
Best used when the bulk of your content consists of elements that can be viewed individually	
These tools are best to use for those projects, wherein the information flow can be directed from beginning to end much like the movies.	

- ii) State the type of authoring tool used in **FIGURE 2**.

[1 mark]

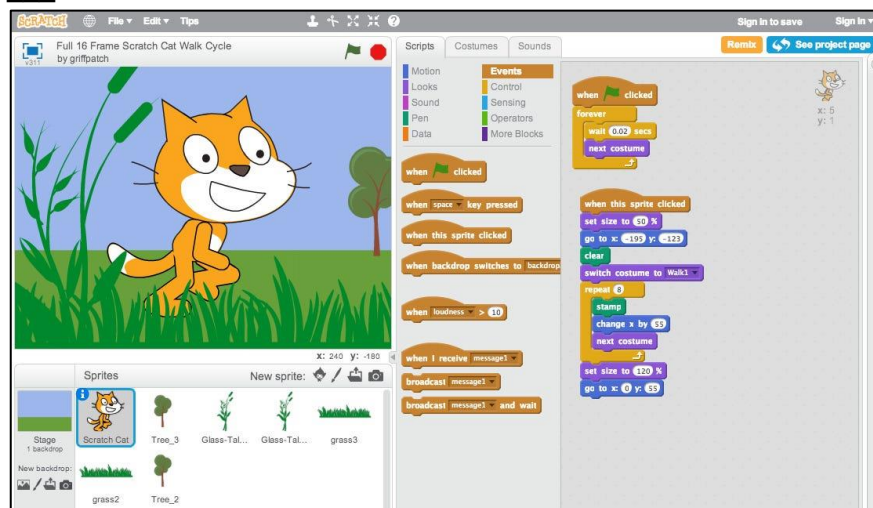


FIGURE 2

- iii) Describe **one (1)** limitation of the type of authoring tool mentioned in question 2(b)(ii).

[2 marks]

- c) Physical devices can be utilized to the distribution of multimedia materials to users. When bandwidth is restricted or when massive files need to be delivered, this technique is frequently employed.
- i) The above statement refers to the one of techniques for distributing multimedia content. What is that technique?

[1 mark]

- ii) List down **two (2)** examples of physical devices that can be used to deliver multimedia content using the technique mentioned in **2(c)(i)**.

[2 marks]

- iii) State **one (1)** disadvantage of using the above technique? Justify your answer.

[2 marks]

3. A small retail business, "Gadgetzs," has been managing its inventory, sales, and customer records using basic file processing systems. As the business expands, the company begins to face challenges in ensuring data consistency, security, and efficient querying. The manager, Shuib, is considering transitioning to a more sophisticated system, a **Database Management System (DBMS)**, to handle the growing complexity.

- a) Differentiate file processing system with database system based on the criteria given.

[4 marks]

	File Processing System	Database System
Data Redundancy		
Data Sharing		

- b) Describe **one (1)** advantage of using database systems.

[2 marks]

4. Databases are crucial in various industries to manage, organize, and retrieve structured data for efficient business operations.

- a) Elisa is the database administrator for a large e-commerce platform that handles multiple customer orders, products, and supplier information. One of the primary responsibilities is ensuring the accuracy and reliability of customer information, inventory details, and transaction records.

Explain the concept of **data integrity**.

[2 marks]

- b) Identify the most suitable **database object** based on the description given below.

[2 marks]

Description	Database Object
Design a structured format that stores each product's details, including ID, name, price, and stock quantity	
Generate a printable document that shows the total sales and revenue generated each month.	

5. In a business setting, IT operations are necessary round-the-clock, highly available networks and services are crucial to guarantee the network is always up and running smoothly and free from interruptions or potential outages.

a) Explain how Application Monitoring System works.

[2 marks]

b) Aside from the use mentioned in part 5(a), state two additional applications of the network.

[2 marks]

c) State the type of wired transmission media based on given usage situation.

[4 marks]

Usage situation	Type of wired transmission media
A user requires a stable and fast internet connection at home from the internet service provider to support multiple devices for streaming, gaming, and remote work.	
A company needs to set up a local area network (LAN) to enable efficient communication and data transfer between various departments in a multi-story office building.	
A small business setting up a security camera system throughout its office. They need reliable, interference-resistant connections to ensure that the video feeds from the cameras are clear and uninterrupted to carry high-frequency signals over longer distances without significant degradation.	
A major hospital needs a high-speed, high-capacity network to support its vast array of medical devices, patient records, and communication systems. It must be able to transmit data at incredibly high speeds, and immune to electromagnetic interference.	

d) State the **type of wireless transmission** media based on the given characteristics.

[4 marks]

Characteristics	Type of wireless transmission media
Base stations will serve a region where it will transmit and receive high frequency radio waves, facilitating communication between your device and the broader network.	
Transmission that uses high-frequency radio waves to transmit data over long distances. These signals can travel through the atmosphere and are commonly used for point-to-point communication, such as between cellular towers or for satellite uplinks.	
Involves sending signals from a ground station to a satellite in orbit. The satellite then relays these signals back to another ground station, allowing for communication over vast distances	
Transmissions work over short distances and require a line of sight between the sending and receiving devices. Think of it as an invisible light beam that carries your data, allowing devices to communicate without cables.	

e) Identify the communication device based on the given function.

[6 marks]

Function	Communication device
This device is translating data from your computer into signals that can travel over the network and vice versa that is usually attached to the computer.	
It directs data between your home or business network and the wider internet, ensuring information gets to the right destination.	
Device that translates digital data from your computer into analog signals that can travel over telephone lines, cable systems, or satellite connections—and vice versa.	

A signal boost for your network. Its job is to take weaker or fading signals and amplify them so they can travel longer distances without losing strength.	
A bridge between your wired network and your wireless devices. It allows Wi-Fi devices like phones, laptops, and tablets to connect to your network without needing physical cables.	
A device that links multiple devices on a network, allowing them to communicate directly and efficiently with each other.	

6. With billions of websites online today, there is a lot of information on the Internet. Jozlyn wonders what kind of information about her can be found in the internet. She typed her name in the search bar and start searching.

- a) i) Identify the most suitable type of **search tool** Jozlyn use to get overview of herself.

[1 mark]

- ii) Describe the type of search tool mentioned above.

[2 marks]

- b) A search strategy is an organized combination of keywords, phrases, subject headings, and limiters used to search a database. Identify the type of search strategies based on the description given:

[3 marks]

Description	Type of search strategies
You are looking for information on Java programming, but you want to exclude results that are related to coffee.	
Finding the article on internet of thing in Journal of big Data.	
Using quotation marks around the phrase ensures that the search engine looks for the exact phrase "internet of things" rather than individual words.	

- c) Web activities are versatile, capable of showing content, product listings, or groupings. The specific action chosen during creation dictates the type of information displayed.

[8 marks]

Scenario	web activities	Example
Lisa always chats her mother whenever she miss her		

Khairul amin manage to all his “Sambal Podeh” within five minutes after he launch the product online.		
Reena shares her personal content, interests, activities, backgrounds or real-life connections.		
Iman completed the 2 chapters online course in order to get a certificate.		

- d) Aisyah is concerned about **online privacy** as websites track her activity showing personalized ads. She fears that companies and third parties may monitor her behaviour across multiple devices without her consent.

i) Based on the above situation, give one (1) example on how websites and third parties can track her activity? **[1 mark]**

ii) How can Aisyah protect her privacy according to your answer in question 6(d)(i)?

[2 marks]

iii) In a country with strict censorship laws, the government forces social media platforms to remove posts that criticize its policies. From your point of view, what is the ethical concern of implementing such censorship from the government? **[1 mark]**

- e) IoT is essential to the 4IR by providing connectivity between devices, enabling real-time data collection, exchange, and analysis. This data drives automation, process optimization, and better decision-making, which are key to the 4IR.

Explain the challenges of integrating the Internet of Things (IoT) within the context of the Fourth Industrial Revolution (4IR)?

[2 marks]

END OF QUESTIONS